

AI-POWERED CAD ANALYSIS REPORT

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Geometry to Process to Cost · Fully AI-reasoned

47

Automotive Fuel Tank (blow-moulded plastic, barrier multilayer)

1528 × 658 × 594 mm · 10595.9 cm³ · AI 28.609 kg / Steel 83.178 kg

Manufacturability: 47/100 · Confidence: High

§1 — Geometry & Part Summary

Bounding Box	1528.3 × 657.6 × 593.7 mm	Surface Area	47571.3 cm ²
Volume	10595.90 cm ³	Weight (AI)	28.609 kg
Weight (Steel)	83.178 kg	Weight (Plastic)	11.126 kg

§2 — Detected Features

Feature Type	Count	Significance	Description
Hollow shell / low fill ratio	1	High	Fill ratio 0.0178 confirms thin-wall hollow blow-moulded body, not a solid casting
Wall thickness variation	29	High	Mean 5.04 mm, max 18.34 mm at pinch-weld/boss regions, $\bar{A} = 2$ EBM parison programming with local pinch-off thickening
Undercuts / side features	753	High	753 undercut faces — brackets, straps, filler neck bosses, baffle mounts requiring split-mould or hand-loaded inserts
Bosses/necks (cylindrical, r"e 3 0 m m)	12	Medium	Filler neck, vent, and mounting boss features consistent with fuel tank fittings
Threaded features	1	Medium	Threaded fitting(s) for filler cap / sender unit, likely moulded-in or post-tapped inserts

§3 — Material Analysis

AI-suggested

Primary Material:

HDPE (High-Density Polyethylene, fuel-tank grade) (70% confidence)

Large hollow automotive fuel tank geometry (1.5m x 0.66m x 0.59m), thick pinch-weld sections, and low fill ratio are classic signatures of extrusion blow moulded HDPE fuel tanks. HDPE with a 6-layer HDPE/tie/EVOH/tie/HDPE barrier construction is the industry-standard resin system for automotive fuel tanks due to chemical resistance, impact toughness, and permeation control. PP and PET are not used for fuel tanks (PET used for beverage bottles via SBM, PP for chemical drums/small IBM bottles).

Alternatives:

PP Homopolymer (MFI 12) (15%) · PA6 (barrier co-extrusion layer only) (10%)

§4 — Process Recommendations

Process	Commodity	Confidence	Cycle Time (hr)	Reasoning
Extrusion Blow Moulding (EBM)	blow_moulding	95%	0.0417	USER-FORCED process. Large single-cavity hollow tank with variable wall (thick pinch welds up to 18.34mm), size >1.5m, and 753 undercut/boss features matches continuous-extrusion parison blow moulding with a split (clam-shell) mould and moving pinch/weld inserts — not IBM (too large, too many undercuts for precision small-bottle process) and not SBM (SBM is stretch-oriented PET bottle process, unsuitable for non-round barrier tanks).

§5 — Manufacturability Risks (Score: 47/100)

Severity	Feature / Area	Description	Recommended Action
High	753 undercut faces	Numerous undercuts (mounting straps, baffle bosses, fitting bosses) require split-mould pinch inserts, moving cores, or post-moulding drilling/welding of brackets	Consolidate undercuts onto the parting line where possible; use retractable core pins for deep bosses; review brackets for post-assembly (welded studs) instead of moulded-in undercuts
High	Wall thickness variation (4.15–18.34mm)	Thick pinch-weld/boss zones (up to 18.34mm, 3.6x the mean) create local hot spots increasing cooling time, sink, and residual stress at weld lines — a critical fuel-tank leak risk area	Programme parison wall profiling (die/mandrel gap control) to even out thickness; add local cooling in pinch zones; specify weld-line minimum thickness per fuel-tank permeation standard
Medium	Large single-cavity size (1528mm)	Very large parison length increases sag/drawdown risk, leading to non-uniform wall distribution over the length of the tank	Use accumulator-head extrusion with programmed die gap (parison programming) rather than continuous extrusion to control sag
Medium	384 zero-draft faces	Zero-draft flat/vertical panels increase mould release friction risk and can cause local thinning during blow-up	Add minimum 1–2° draft on large flat panels where cosmetically acceptable

§6 — DFM Issues (blow_moulding)

Severity	Area	Description	Impact	Fix
High	Pinch-weld thickness (max 18.34mm)	Local wall thickness at pinch/weld lines is 3.6x the mean wall, creating slow-cooling hot spots and potential weak weld-line permeation paths critical for a fuel tank	Extended cycle time (cooling-dominated), risk of field leaks/warranty failures, extra QC (leak/burst testing)	Refine parison programming (die/mandrel profile) to reduce pinch-off excess material; add localized mould cooling channels at weld zones; specify minimum weld-line thickness per SAE/ISO fuel tank permeation spec
High	753 undercut / boss features	High undercut count for brackets, filler neck, vent and sender bosses requires numerous mould side-actions or pinch-off inserts	Increased mould complexity and cost, longer mould build lead time, risk of flash/witness lines at inserts	Move non-critical brackets to post-mould welded/bonded studs; group undercuts along the parting plane where feasible
Medium	6-layer barrier co-extrusion	HDPE/tie/EVOH/tie/HDPE barrier structure requires a precision 6-layer die head with tight layer-thickness control to guarantee permeation compliance	Higher tooling (die head) cost and tighter process control (higher scrap if layer ratios drift)	Specify layer-thickness monitoring (ultrasonic gauge) and maintain EVOH layer "e" minimum per permeation regulation
Medium	Large parison sag over 1.5m length	Long, heavy parison for a part of this length is prone to gravitational	Non-uniform wall thickness lowers burst strength and increases	Use accumulator-head extrusion with programmed die gap

Severity	Area	Description	Impact	Fix
		sag/drawdown, producing thin walls at the top and thick at the bottom	scrap/rework	opening timed to counter sag, and orient parting line to minimize drop height
Low	384 zero-draft faces	Flat, near-vertical panels with no draft increase mould release drag during ejection of the blown part while still warm	Minor surface scuffing/witness marks, marginally longer cycle for careful ejection	Add 1-2° draft to large flat panels not constrained by packaging envelope

\$7 — AI Indicative Cost Range (model opinion — not engine-calculated)

OPTIMISTIC	MOST LIKELY	CONSERVATIVE
£19.50	£27.00	£38.00

Net Weight	10.172 kg	Material	mat-hdpe-fuel-coex
Recommended Process	blow_moulding	Cycle Time	0.0417 hr/part
Setup Time	0.750 hr	Operations	0 ops

Operations Detail

Process-Specific Parameters

Casting Subtype	sand
Die/Mould Cost	£50,000
Die Life	500 shots
Cavities	1
Yield	85.0%
Flash Weight	0.000 kg
Yield	0.0%
Die Cost	£0
Strokes	0
Cavities	0
Wall Thickness	0 mm
Mould Cost	£0
Mould Life	0 shots
Projected Area	0.0 cm²

§8 — AI Analysis Explanation

This is a large (1.53m x 0.66m x 0.59m) hollow thin-wall shell with a very low fill ratio (0.0178), consistent with an automotive fuel tank. Wall thickness is thin and fairly uniform (mean 5.04mm) except at pinch-weld/boss zones which read up to 18.34mm — a classic EBM parison pinch-off signature, not a cast or machined solid. The user has forced blow_moulding as the process; given the part's role (fuel tank), size, and thick weld zones, Extrusion Blow Moulding (EBM) with a single large clam-shell mould is the correct subtype (vs IBM for small precision bottles or SBM for PET beverage containers). Material was undecided in the geometry package; HDPE with a 6-layer HDPE/EVOH barrier structure was selected as the industry-standard resin for automotive fuel tanks (chemical resistance + permeation control), over PP (used for drums/IBM bottles) and PET (used for SBM beverage bottles, unsuitable for hydrocarbon service). Net part weight was recalculated at HDPE's true density (0.955 g/cm³) rather than the generic 1.05 g/cm³ 'Plastic' figure supplied, giving ~10.12kg net, with an estimated ~2.5kg of trim flash from the continuous parison process (pinch weld waste, typically reground and reused). Mould cost, cavity count, and cycle time (blow/cool/open-close) are rule-of-thumb estimates informed by geometry (size, wall, undercuts) since no OCCT-deterministic blow-moulding parameters were provided beyond wall thickness.

§9 — Analysis Limitations & Assumptions

1. Resin identity (HDPE vs PP vs PET) was not specified in metadata; HDPE assumed based on part function as a fuel tank, industry convention
2. Flash weight, cavity count, blow/cool/open-close times, and mould cost are rule-of-thumb estimates informed by geometry, not measured/deterministic values
3. No matching blow-moulding machineId exists in the provided valid machineId list (all listed IDs are CNC/machining equipment) — estimatedOperations left empty to avoid mis-assigning a machining resource to a blow-moulding process
4. Barrier layer structure (6-layer HDPE/EVOH) assumed from part function (automotive fuel tank) per project guidance, not confirmed from drawing notes

Stage-1 pre-selection: auto (0%) ·